

**An Efficient Single-Channel Quadrature-Phase Learning-Based Approach  
for Angle-of-Arrival Estimation**

**Abstract**

Angle-of-arrival (AoA) estimation using machine learning (ML) can be formulated as a nonlinear function approximation problem; however, its performance is often limited by feature ambiguity and high computational complexity. This paper proposes an efficient single-channel approach for ML-based AoA estimation by exploiting the quadrature-phase (imaginary) component of the normalized spatial covariance matrix as the input feature vector. The proposed feature representation preserves deterministic inter-element phase differences, thereby enabling a unique mapping between the extracted features and the corresponding AoA. Radial basis function neural networks (RBFNN) and support vector regression (SVR) are employed as benchmark learning models to validate the proposed formulation. Simulation results demonstrate that the proposed approach achieves superior estimation accuracy and robustness compared to existing methods, particularly under conditions of reduced array elements, limited data snapshots, and low signal-to-noise ratio (SNR). Furthermore, the proposed feature extraction significantly reduces computational complexity during both training and testing phases, resulting in improved execution efficiency.

**Keywords:** angle-of-arrival estimation, machine learning, quadrature phase, radial basis function neural network, support vector regression.

**1. Introduction**

Angle-of-arrival (AoA) estimation is a fundamental problem in sensor array and multichannel (SAM) signal processing and has been extensively studied over the past several decades. Traditionally, AoA estimation has been primarily associated with defence-related applications such as radio detection and ranging (RADAR) and sound navigation and ranging (SONAR) systems [1], [2]. In recent years, however, AoA estimation has become a key enabling technology in a wide range of civilian applications, including ultrasound imaging, automotive radar for autonomous driving,

microphone array systems for human-machine interaction (e.g., smart speakers and audio zooming), and multiple-input multiple-output (MIMO) antenna arrays for modern wireless communication standards such as IEEE 802.11n/ac, 5G, and beyond [3]. Early approaches to AoA estimation can be broadly classified into classical beamforming methods, maximum likelihood methods, subspace-based methods, and hybrid techniques [4]. Among these, subspace-based algorithms such as multiple signal classification (MUSIC) and estimation of signal parameters via rotational invariance techniques (ESPRIT), along with their variants, are widely regarded as benchmark methods for evaluating the advances in AoA estimation [5], [6]. Later, compressive sensing (CS)-based approaches have gained attention for AoA estimation under sparse and underdetermined scenarios [7], [8]. Despite their effectiveness, these approaches often suffer from high computational complexity and sensitivity to model mismatch, particularly in low signal-to-noise ratio (SNR) or limited snapshot regimes.

Motivated by the need for reduced computational complexity and improved robustness, machine learning (ML) techniques have been increasingly explored for AoA estimation [9]. In ML-based formulations, AoA estimation is treated as a nonlinear function approximation problem that maps the received array measurements to the corresponding source directions (AoA). This mapping can be learned using supervised learning models such as radial basis function neural networks (RBFNNs) [10]–[13] and support vector regression (SVR) [14]–[17]. RBFNNs employ a single hidden layer with radial basis activation functions; however, their performance strongly depends on the selection of optimal number of hidden neurons, which often requires heuristic tuning [14]. In contrast, SVR-based approaches are generally easier to implement and exhibit superior generalization performance with comparatively lower computational complexity, making them one of the most widely adopted ML techniques for AoA estimation [16], [17].

In existing RBFNN- and SVR-based AoA estimation frameworks, the spatial covariance matrix (SCM) of the received array data is commonly vectorized and used as the input feature for training the learning models. Owing to its Hermitian symmetry, the SCM contains redundant information, and only its upper (or lower) triangular elements are sufficient to capture the spatial characteristics related to AoA [11]. However, the resulting feature vectors are complex-valued, whereas conventional ML models operate on real-valued data [18], [19]. A common strategy to address this issue is the dual-channel formulation, in which the real and imaginary parts of the complex-valued features are treated as separate real-valued inputs [20]. While straightforward,

this approach ignores the intrinsic correlation between the real and imaginary components, effectively doubles the feature dimensionality, and may degrade the generalization capability and robustness of the learned models.

To overcome these limitations, this paper proposes an efficient single-quadrature-channel formulation for ML-based AoA estimation. The proposed approach exploits only the quadrature-phase (imaginary) component of the normalized spatial covariance matrix while discarding the principal diagonal elements, which contain only per-element power information and do not contribute to AoA discrimination. By leveraging deterministic inter-element phase differences, the proposed feature representation preserves the angular information required for AoA estimation while significantly reducing feature dimensionality and computational complexity.

The main contributions of this paper are summarized as follows:

1. A novel single-quadrature-channel feature representation based on the imaginary part of the normalized spatial covariance matrix is proposed for ML-based AoA estimation.
2. Uniqueness of the feature-to-AoA mapping is analytically demonstrated, enabling effective and reliable learning using standard ML models.
3. Reduced computational complexity is achieved during both training and testing phases compared to existing dual-channel ML-based AoA estimation approaches.

## 2. Data Model

Consider  $M$  far-field (plane wave) targets from direction angles  $\Theta = \{\theta_m\}_{m=1}^M$  impinging on an  $N$ -element uniform linear array (ULA) with a field-of-view (FOV) such that  $\theta_m \in (-90^\circ, +90^\circ)$  as shown in Fig.1. Then, the output of the ULA in the baseband,  $\mathbf{x}(t) \in \mathbb{C}^{N \times 1}$  can be expressed as

$$\mathbf{x}(t) = \sum_{m=1}^M \mathbf{a}(\theta_m) s_m(t) + \mathbf{n}(t), \quad t = 1, \dots, T \quad (1)$$

where  $T$  represents the number of data snapshots.  $\mathbf{a}(\theta_m) \in \mathbb{C}^{N \times 1}$  denotes the array directional vector associated with the  $m^{\text{th}}$  target direction angle  $\theta_m \in \mathbb{R}^1$  is given by

$$\mathbf{a}(\theta_m) = [1, e^{-jk \sin(\theta_m)}, \dots, e^{-jk(N-1)\sin(\theta_m)}]^T \quad (2)$$

where  $k = \frac{2\pi d}{\lambda}$ ,  $d = \frac{\lambda}{2}$  is the inter-element spacing of the ULA,  $\lambda$  is the wavelength of the target signal.  $s_m(t) \in \mathbb{C}^{N \times 1}$  denotes the signal amplitude of the  $m^{\text{th}}$  target.  $\mathbf{n}(t) \in \mathbb{C}^{N \times 1}$

represents the additive white Gaussian noise with zero mean and covariance  $\mathbb{E}\{\mathbf{n}(t)\mathbf{n}^H(t)\} = \sigma_n^2 \mathbf{I}_N$ , where  $\sigma_n^2$  is the noise power and  $\mathbf{I}_N$  denotes the  $N \times N$  identity matrix [11].

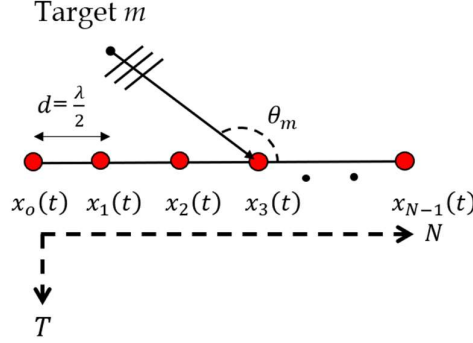


Fig.1. Configuration of the data model

### 3. Proposed Formulation

This section presents the formulation of the proposed AoA estimation approach using quadrature phase information extracted from the spatial covariance matrix (SCM) and its complexity analysis

#### 3.1 Data Processing and Quadrature Feature Extraction

The proposed approach aims to approximate the unknown nonlinear mapping between the quadrature-phase information of the spatial covariance matrix ( $\mathfrak{I}(\mathbf{R})$ ) and the corresponding angle-of-arrival ( $\theta$ ) of the incoming target. This mapping is represented as  $\hat{F}: \mathfrak{I}(\mathbf{R}) \rightarrow \theta$  and is learned using machine learning (ML) models such as radial basis function neural networks (RBFNN) and support vector regression (SVR).

The SCM of the received array data  $\mathbf{x}(t) \in \mathbb{C}^{N \times 1}$  is defined as

$$\mathbf{R} = \mathbb{E}\{\mathbf{x}(t)\mathbf{x}^H(t)\} \quad (3)$$

In practice, the SCM ( $\mathbf{R}$ ) is typically unknown and estimated from the received data  $\mathbf{x}(t)$ [16]. The estimated SCM for  $T$  snapshots of data  $\mathbf{x}(t)$  can be expressed as

$$\hat{\mathbf{R}} = \frac{1}{T} \sum_{t=1}^T \mathbf{x}(t) \mathbf{x}^H(t) = \begin{bmatrix} r_{0,0} & r_{0,1} & \cdots & r_{0,N-1} \\ r_{1,0} & r_{1,1} & \cdots & r_{1,N-1} \\ \vdots & \vdots & \ddots & \vdots \\ r_{N-1,0} & \cdots & \cdots & r_{N-1,N-1} \end{bmatrix} \quad (4)$$

where  $r_{p,q}$  represents the covariance between the  $p^{th}$  and  $q^{th}$  array element's received data and it is given by

$$r_{p,q} = \mathbb{E}\{x_p(t)x_q^*(t)\} = \frac{1}{T} \sum_{t=1}^T x_p(t)x_q^*(t), p, q = 0, 1, \dots, N-1 \quad (5)$$

The diagonal elements  $r_{p,p}$  correspond to the received power at each array element, while the off-diagonal elements capture the relative phase differences induced by the incident wavefronts. Due to the Hermitian symmetry of  $\hat{\mathbf{R}}$ , only the off-diagonal elements are required to characterize the AoA information. To derive the quadrature-phase feature, consider a narrowband plane wave impinging on a uniform linear array (ULA). The received signal at the  $p$ -th element is given by

$$x_p(t) = s(t)e^{-j2\pi(p-1)\frac{d}{\lambda}\sin\theta} + n_p(t) \quad (6)$$

where  $s(t)$  is the source signal,  $d$  is the inter-element spacing,  $\lambda$  is the wavelength, and  $n_p(t)$  is additive noise. Substituting (6) into (5) and assuming independence between signal and noise yields

$$r_{p,q} = \sigma_s^2 e^{-j2\pi(p-q)\frac{d}{\lambda}\sin\theta} + \sigma_n^2 \delta_{p,q} \quad (7)$$

In Equation (7), the only term dependent on  $\theta$  is the exponential phase term; the signal and noise powers do not encode AoA information. Hence, all AoA information is contained in the inter-sensor phase differences.

For  $p \neq q$ , the noise term vanishes and (7) reduces to

$$r_{p,q} = \sigma_s^2 e^{-j2\pi(p-q)\frac{d}{\lambda}\sin\theta} \quad (8)$$

To remove amplitude dependence and retain only phase information, the SCM is normalized element-wise as

$$u_{p,q} = \frac{r_{p,q}}{\sqrt{r_{p,p}r_{q,q}}} = e^{-j2\pi(p-q)\frac{d}{\lambda}\sin\theta} \quad (9)$$

Applying Euler's identity,

$$u_{p,q} = \cos\left(2\pi(p-q)\frac{d}{\lambda}\sin\theta\right) - j\sin\left(2\pi(p-q)\frac{d}{\lambda}\sin\theta\right) \quad (10)$$

The real part is deterministically related to the imaginary part and does not provide additional independent AoA information. Therefore, only the quadrature (imaginary) component is retained:

$$\Im(u_{p,q}) = -\sin\left(2\pi(p-q)\frac{d}{\lambda}\sin\theta\right) \quad (11)$$

For  $d = \lambda/2$ , Equation (11) simplifies to

$$\Im(u_{p,q}) = -\sin(\pi(p-q)\sin\theta) \quad (12)$$

For feature vector construction, the quadrature feature associated with the array element pair  $(p, q)$  is defined as

$$v_{p,q} = \Im(u_{p,q}), p < q \quad (13)$$

Since  $u_{p,p} = 1$ , the principal diagonal elements are excluded. The final feature vector is obtained by stacking all distinct quadrature components:

$$\mathbf{v} = [v_{1,2}, v_{1,3}, \dots, v_{N-1,N}]^T \in \mathbb{R}^D; \quad D = \left(\frac{N(N+1)}{2}\right) - N \quad (14)$$

*Proposition for Uniqueness of Quadrature-Phase Mapping: The mapping  $\theta \mapsto \mathbf{v} = [v_{1,2}, v_{1,3}, \dots, v_{N-1,N}]^T$  where  $v_{p,q} = \Im(u_{p,q}), p < q$  is injective, i.e., each AoA  $\theta \in (-90^\circ, 90^\circ)$  corresponds to a unique quadrature vector  $\mathbf{v}$ .*

*Proof: Over the interval  $\theta \in (-90^\circ, 90^\circ)$ , the function  $\sin \theta$  is strictly monotonic.*

*Assume that two distinct angles  $\theta_1, \theta_2 \in (-90^\circ, 90^\circ)$  produce the same quadrature vector:  $v_{p,q}(\theta_1) = v_{p,q}(\theta_2), \forall p < q$ . In particular, for the adjacent sensor pair ( $p - q = 1$ ),  $\sin(\pi \sin \theta_1) = \sin(\pi \sin \theta_2)$ . Let  $a = \pi \sin \theta_1, b = \pi \sin \theta_2$ . Since  $\sin \theta \in (-1, 1)$ , we have  $a, b \in (-\pi, \pi)$ . From the identity  $\sin a = \sin b \Rightarrow a = b + 2k\pi$  or  $a = \pi - b + 2k\pi, k \in \mathbb{Z}$ , and because  $a, b \in (-\pi, \pi)$ , neither of the periodic alternatives can remain within this interval unless  $a = b$ . Hence,  $\pi \sin \theta_1 = \pi \sin \theta_2$ , which gives  $\sin \theta_1 = \sin \theta_2$ . Since  $\sin \theta$  is strictly monotonic on  $(-90^\circ, 90^\circ)$ , it follows that  $\theta_1 = \theta_2$ . This contradicts the assumption that  $\theta_1 \neq \theta_2$ . Therefore, the mapping is injective.*

Within the array field-of-view  $\theta \in (-90^\circ, 90^\circ)$ , the quadrature component provides a unique and deterministic mapping to the AoA. Consequently, only the quadrature-phase information is retained for feature construction. This single-quadrature-channel representation significantly reduces feature dimensionality while preserving all unique AoA information, thereby providing an efficient and robust input for data-driven AoA estimation.

### 3.2 Training Phase

The training phase is performed offline to learn the nonlinear mapping between the proposed quadrature-phase feature vector ( $\mathbf{v}$ ) and the corresponding angle-of-arrival ( $\theta$ ). The AoA estimation model is expressed as  $\hat{F}: \mathbf{v} \rightarrow \theta$ . A set of  $L$  labeled training pairs is generated as  $\mathcal{D} = \{(\mathbf{v}_l, \theta_l)\}_{l=1}^L$  where the angles  $\theta_l$  are uniformly sampled over the angular range  $[-90^\circ, +90^\circ]$  with angular separation  $\Delta\theta$ , such that  $\theta_l = -90^\circ + (l-1)\Delta\theta, l = 1, 2, \dots, L$ . For each  $\theta_l$ , the spatial covariance matrix  $\hat{\mathbf{R}}^{(l)}$  is computed from measured array data, and the corresponding quadrature feature vector  $\mathbf{v}_l$  is extracted using (14). These training pairs are used to learn  $\hat{F}(\cdot)$  using standard

machine learning regression models. It is emphasized that the contribution of this work lies in the quadrature-phase feature representation, while RBFNN and SVR are employed as benchmark learning models to validate the effectiveness of the proposed formulation.

### 3.2.1 RBFNN-Based AoA Estimation

In the RBFNN-based approach, the nonlinear mapping between the feature vector and the AoA is approximated using a radial basis function neural network with Gaussian kernel. The RBFNN output is expressed as

$$\hat{F}_{\text{RBFNN}}(\mathbf{v}) = \sum_{i=1}^C w_i \exp \left( -\frac{\|\mathbf{v} - \eta_i\|^2}{\rho^2} \right) \quad (15)$$

where  $C$  denotes the number of radial basis functions,  $\eta_i$  are the centers,  $\rho$  is the kernel width, and  $w_i$  are the synaptic weights.

The network parameters are learned by minimizing the mean squared error between the predicted AoA and the true AoA over the training set using standard least-squares or recursive least-squares (RLS) techniques. The number of centers  $C$  and the kernel width  $\rho$  are selected empirically to achieve optimal performance [12].

### 3.2.2 SVR-Based AoA Estimation

Support vector regression (SVR) is employed as an alternative learning model due to its strong generalization capability and robustness to overfitting. The SVR-based AoA estimator is given by

$$\hat{F}_{\text{SVR}}(\mathbf{v}) = \sum_{i=1}^L (\alpha_i - \alpha_i^*) \mathcal{K}(\mathbf{v}_i, \mathbf{v}) + b \quad (16)$$

where  $\alpha_i$  and  $\alpha_i^*$  are Lagrange multipliers,  $b$  is the bias term, and  $\mathcal{K}(\cdot, \cdot)$  denotes the Gaussian kernel  $\mathcal{K}(\mathbf{v}_i, \mathbf{v}) = \exp(-\gamma \|\mathbf{v} - \mathbf{v}_i\|^2)$ .

The SVR model parameters are obtained by minimizing the regularized risk function

$$R_{\text{reg}} = \sum_{i=1}^L l_{\varepsilon}(\mathbf{v}_i, \theta_i) + \frac{1}{2} \|\mathbf{w}\|^2 \quad (17)$$

where  $l_{\varepsilon}(\cdot)$  is the  $\varepsilon$ -insensitive loss function,  $\gamma$  is the kernel parameter,  $\varepsilon$  defines the error tolerance, and  $\zeta$  is the regularization constant. The hyperparameters  $\{\gamma, \varepsilon, \zeta\}$

are selected using standard tuning strategies to ensure optimal generalization performance [15], [21].

### 3.3 Testing Phase

In the testing (estimation) phase, the trained AoA estimation models  $\hat{F}_{\text{RBFNN}}(\mathbf{v})$  and  $\hat{F}_{\text{SVR}}(\mathbf{v})$  are utilized to estimate the unknown  $M$  AoA's  $\{\theta_m\}_{m=1}^M$  of the incoming targets. These estimates are computed as

$$\{\hat{\theta}_m\}_{m=1}^M = \hat{F}(\mathbf{v}^m) \quad (18)$$

where  $\mathbf{v}^m$  represents the feature vector of  $m^{\text{th}}$  target and it is computed from the array SCM ( $\hat{\mathbf{R}}$ ) using Equation (14).

### 3.4 Complexity Analysis

The computational complexity of the proposed approach is analyzed in terms of the number of complex multiplications required during the training and testing phases. In both phases, the total complexity consists of two components: (1) SCM computation complexity, required to extract the input features. (2) Model complexity, associated with the learning and inference stages of the machine learning models. The computation of the SCM for an array with  $N$  elements and  $T$  snapshots require  $O(N^2T)$  complex multiplications for both the existing and proposed approaches. The difference in computational complexity arises from the dimensionality of the feature vector used to train and test the ML models. The proposed single-channel formulation utilizes only the quadrature-phase components of the normalized off-diagonal elements. This reduction directly lowers the computational cost of both RBFNN- and SVR-based AoA estimation. Table 1 summarizes the computational complexity of the existing and proposed approaches for both training and testing phases.

**Table 1: Computational Complexity Comparison**

| Approach | Model              | Training Complexity                                                      | Testing Complexity                                                          |
|----------|--------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Existing | RBFNN<br>[10]-[13] | $O(N^2T + CLN(N+1))$                                                     | $O(N^2T + CN(N+1))$                                                         |
| Existing | SVR<br>[14], [15]  | $O(N^2T + L^2N(N+1))$                                                    | $O(N^2T + n_{sv}N(N+1))$                                                    |
| Proposed | RBFNN              | $O\left(N^2T + CL\left(\left(\frac{N(N+1)}{2}\right) - N\right)\right)$  | $O\left(N^2T + C\left(\left(\frac{N(N+1)}{2}\right) - N\right)\right)$      |
| Proposed | SVR                | $O\left(N^2T + L^2\left(\left(\frac{N(N+1)}{2}\right) - N\right)\right)$ | $O\left(N^2T + n_{sv}\left(\left(\frac{N(N+1)}{2}\right) - N\right)\right)$ |



where  $C$  is the number of RBF centers,  $L$  is the number of training samples, and  $n_{sv}$  denotes the number of support vectors in SVR. From Table 1, it is evident that the proposed single-channel formulation significantly reduces computational complexity in both training and testing phases.

#### 4. Simulation Results

This section presents the simulation results of the proposed approach in comparison with existing ML-based AoA estimation methods using RBFNN [10]–[13] and SVR [14], [15]. Performance is evaluated in terms of estimation reliability, estimation accuracy, computational complexity, and execution time.

##### 4.1 Estimation Reliability

The estimation reliability of the proposed approach is evaluated under two conditions: noise-free ( $\sigma_n^2 = 0$ ) and noisy ( $\sigma_n^2 = 0.1$ , corresponding to an SNR of 10 dB). The simulations consider a single target ( $M = 1$ ), with  $N = 5$  array elements and  $T = 1000$  data snapshots. For training,  $L = 162$  randomly generated samples distributed over the angular range  $-90^\circ$  to  $+90^\circ$  with a resolution of  $\Delta\theta = 1^\circ$  are used. An additional 19 angular samples, not included in the training set, are reserved for testing. The hyperparameters used during training are as follows: for the existing approach,  $C = 100$ ,  $\sigma = 0.1$ ,  $\zeta = 100$ ,  $\varepsilon = 1$ ,  $\gamma = 0.5$ . For the proposed approach,  $C = 100$ ,  $\sigma = 1$ ,  $\zeta = 100$ ,  $\varepsilon = 1$ ,  $\gamma = 0.5$ . The selection of hyperparameters is based on training accuracy and is detailed in Appendix A. Figures 2(a)–(d) and 2(e)–(h) show that the proposed approach consistently estimates the true AoAs with higher reliability than the existing approaches for both RBFNN and SVR, under both noise-free and noisy conditions.

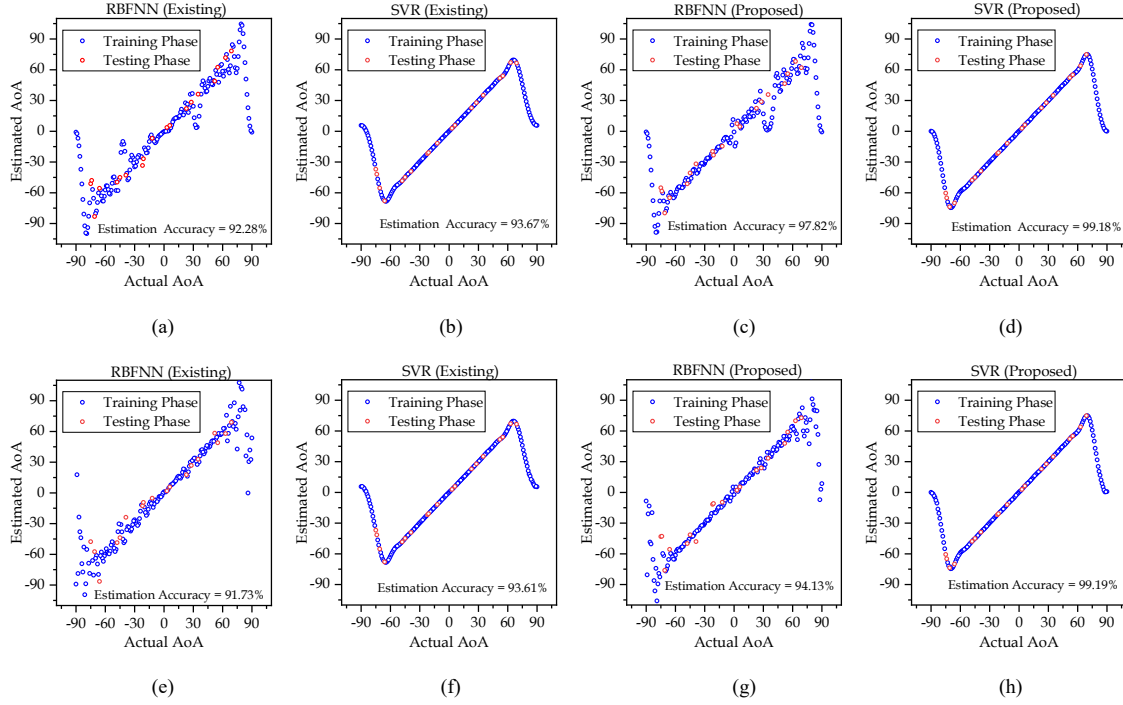


Fig. 2. Reliability comparison: (a)-(d) under  $\sigma_n^2 = 0$ , (e)-(h) under  $\sigma_n^2 = 0.1$

#### 4.2 Estimation Accuracy

Estimation accuracy is evaluated under variations in number of array elements, number of snapshots, and SNR. The impact of the number of array elements on estimation accuracy is evaluated for  $2 \leq N \leq 15$  with fixed  $T = 1000$  and  $\sigma_n^2 = 0.1$  (10 dB SNR). As shown in Fig. 3, the proposed SVR approach achieves superior accuracy compared to the existing SVR approach, even with a small number of array elements. The proposed RBFNN approach also consistently outperforms its existing counterpart as  $N$  increases. The effect of snapshot variation is evaluated for  $10 \leq T \leq 500$  with fixed  $N = 5$  and  $\sigma_n^2 = 0.1$ . As illustrated in Fig. 4, the proposed SVR approach provides improved estimation accuracy across all snapshot values. Additionally, the proposed RBFNN approach demonstrates more stable and accurate performance than the existing RBFNN approach as  $T$  increases. Estimation accuracy is further evaluated for SNR values ranging from  $-20$  dB to  $+20$  dB with fixed  $N = 5$  and  $T = 1000$ . Fig. 5 shows that the proposed SVR approach consistently achieves superior accuracy, particularly in low-SNR regimes, compared to the existing methods.

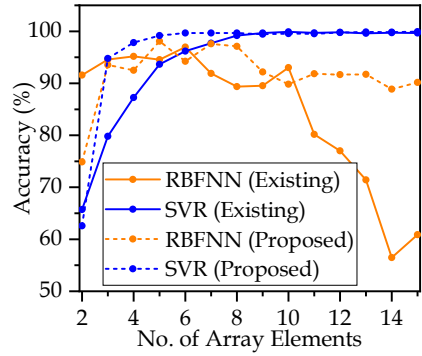


Fig.3. Accuracy comparison versus number of array elements

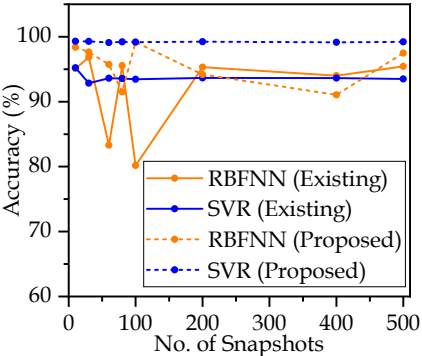


Fig.4. Accuracy comparison versus number of snapshots

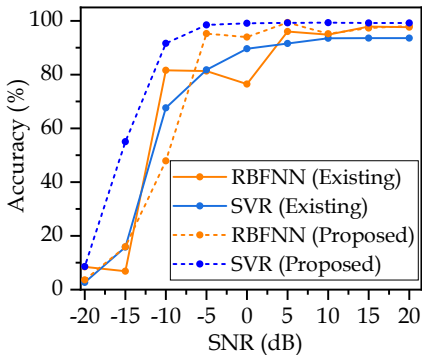


Fig.5. Accuracy comparison versus SNR

4.3 Computational Complexity and Execution Time

The training and testing computational complexities are evaluated for  $3 \leq N \leq 10$  with fixed  $T = 1000$ ,  $C = 100$ , and  $L = 162$ . Figures 6 and 7 show that the proposed SVR approach reduces computational complexity by approximately 50% compared to the existing SVR approach across all values of  $N$ . The proposed RBFNN approach achieves the lowest training complexity; however, the reduction in testing complexity is less pronounced due to the relatively small number of training samples. Execution

time is measured during the testing phase for  $N = 5$  and  $T = 1000$  on a PC equipped with a 1.8 GHz processor and 8 GB RAM. The results are summarized in Table 2, where the proposed approach demonstrates significantly reduced execution time compared to the existing methods.

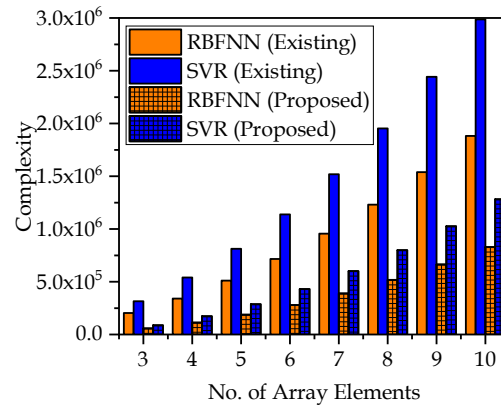


Fig.6. Training complexity comparison versus number of array elements

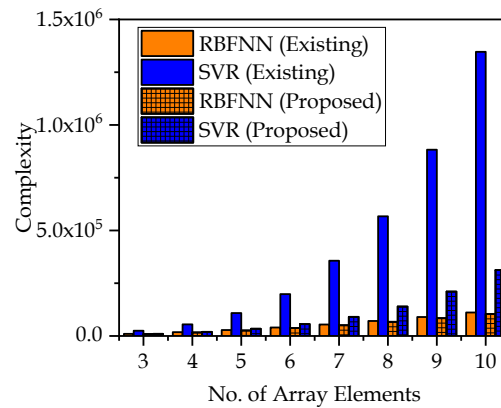


Fig.7. Testing complexity comparison versus number of array elements

**Table 2: Testing Phase Execution Time Comparison**

| Method | Existing Approach (s) | Proposed Approach (s) |
|--------|-----------------------|-----------------------|
| RBFNN  | 0.00152               | 0.00101               |
| SVR    | 0.00802               | 0.00171               |

## 5. Conclusion

This paper presents an efficient single-channel machine learning-based AoA estimation approach using quadrature-phase information extracted from the spatial

covariance matrix. The proposed formulation ensures a unique and deterministic mapping between features and AoA, enabling effective learning using standard ML models. Simulation results demonstrate that the proposed approach consistently outperforms existing methods in terms of estimation reliability and accuracy, particularly under challenging conditions involving fewer array elements, limited snapshots, and low SNR. Moreover, the proposed approach significantly reduces computational complexity and execution time during both training and testing phases, making it well-suited for practical real-time AoA estimation applications.

### Appendix A

In Appendix A, simulation analysis for the selection of hyperparameter values based on the training phase accuracy for different sets of hyperparameters is presented in Fig.A.1, considering  $M = 1$ ,  $N = 5$  and  $T = 1000$  under  $\sigma_n^2 = 0$ .

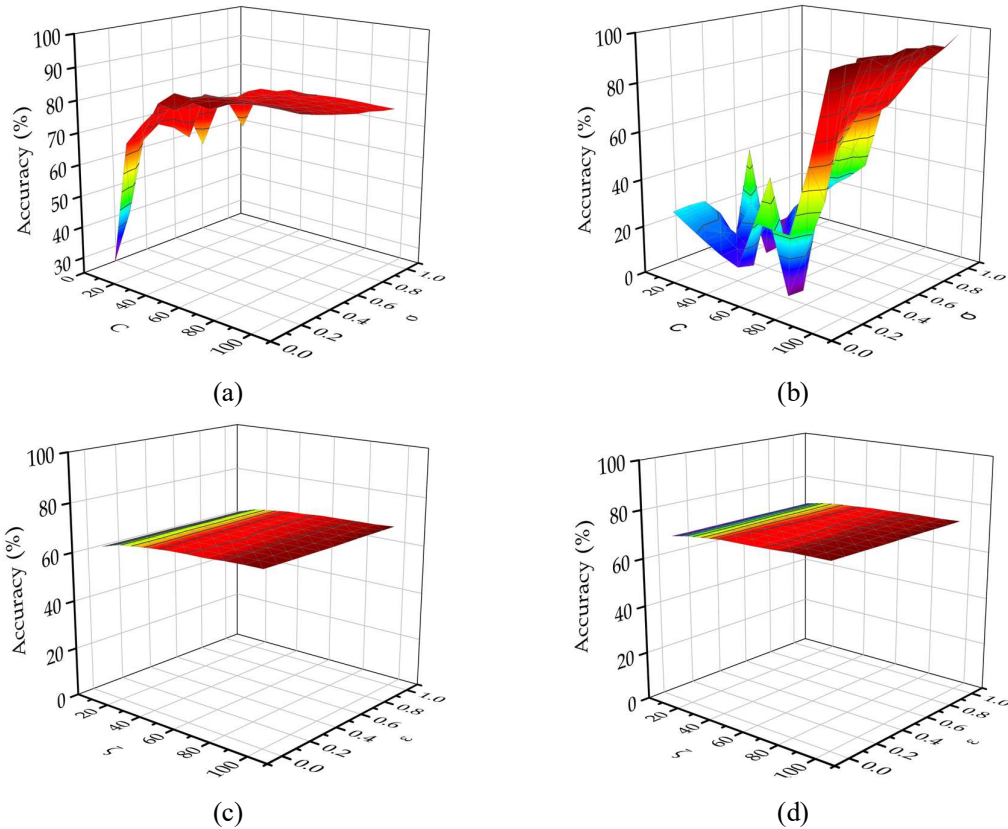


Fig. A.1. Training accuracy versus hyperparameters: (a) RBFNN (existing approach), (b) RBFNN (proposed approach), (c) SVR (existing approach), (d) SVR (proposed approach).

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